Chapter 28

What to Expect From a Pharmacist Preceptor

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Learning Objectives

Objective 28.1 Differentiate the roles of a preceptor and clarify the terms preceptor, mentor, faculty, intern, and resident.
Objective 28.2 Describe the qualities of effective preceptors and responsibilities of students.
Objective 28.3 Detail how to develop a productive student–preceptor relationship.

OBJECTIVE 28.1. DIFFERENTIATE THE ROLES OF A PRECEPTOR AND CLARIFY THE TERMS PRECEPTOR, MENTOR, FACULTY, INTERN, AND RESIDENT

Learning is a process. It begins with a foundation of knowledge and should continue throughout a pharmacist's career. A preceptor is a teacher in a clinical setting who will help students grow from a novice into a competent, professional pharmacist. During experiential education in the pharmacy curriculum, student pharmacists are taught by experienced pharmacist preceptors. Many of these individuals have had some sort of training or orientation to become preceptors and should be motivated by a desire to teach the next generation of pharmacists. To have a clear understanding of what a student expects from his/her preceptor, it is important to differentiate the various roles and terms related to being a preceptor.

A *faculty member* is an educator who is employed by a college, school, or university. Responsibilities of faculty members are threefold—research, service, and teaching. Typically, most faculty members have areas of expertise which serve as the foundation for their research and quest for new knowledge. Additionally, faculty members provide service to the greater university and their community (e.g., committee work, serving as a reviewer for a journal, patient care service). A faculty member may teach in a classroom or laboratory as well as in his/her pharmacy practice site. While teaching in a pharmacy practice site, the faculty member would also be a preceptor. A *preceptor* is a pharmacist and educator (or in rare cases another healthcare provider) who provides the practice-based instruction to students at a pharmacy practice site. A preceptor may be employed by a college or school of pharmacy or may be employed by a pharmacy or health system. An *intern* is typically a student or trainee who works with or without payment to gain knowledge and experience related to the practice of pharmacy. And, a *pharmacy resident* is a licensed practitioner who is training under the supervision of an experienced pharmacist preceptor in a specific area of pharmacy practice.

During a student's numerous interactions with a preceptor, because of this relationship (i.e., preceptorship), the preceptor may also become a trusted *mentor and advisor*. Oftentimes, these mentor relationships may extend for the student's career and can provide a mechanism for the learner to receive advice and guidance for a lifetime. This highlights the important facets to consider when communicating and interacting with a preceptor.

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Practice Scenario

You are assigned to complete a practice-based experience in an area of pharmacy you plan to pursue after graduation and licensure. During the experience, you show tremendous growth and perform at a very high level, earning the praise of many of your preceptors. Following the experience you continue to ask your preceptor for advice and guidance regarding your career.

- During the experience, what is your relationship to the preceptor? *During the experience you are in a preceptor–student type of relationship. The preceptor is involved in teaching and facilitating learning for you.*
- Following the experience, what is your relationship to the preceptor? *Following the experience, you are beginning to develop more of a mentor–mentee relationship with the preceptor that may extend your career as you look to this person for guidance and advice.*

The Roles of a Preceptor

Preceptors take on many roles as they provide instruction to students such as providing direct instruction, modeling, coaching, and facilitating.[1] Preceptors can utilize all four roles during a given day or during a type of experience. The four preceptor roles align with Miller's pyramid of learning (Figs. 28.1 and 28.2).[2] The base of Miller's pyramid begins with



FIGURE 28.1 Miller's pyramid of assessment of clinical competence.[2] *Used with permission: Miller GE. The assessment of clinical skills/competence/performance.* Acad Med. *1990;65:563–567. https://insights.ovid.com/pubmed?pmid=2400509.*

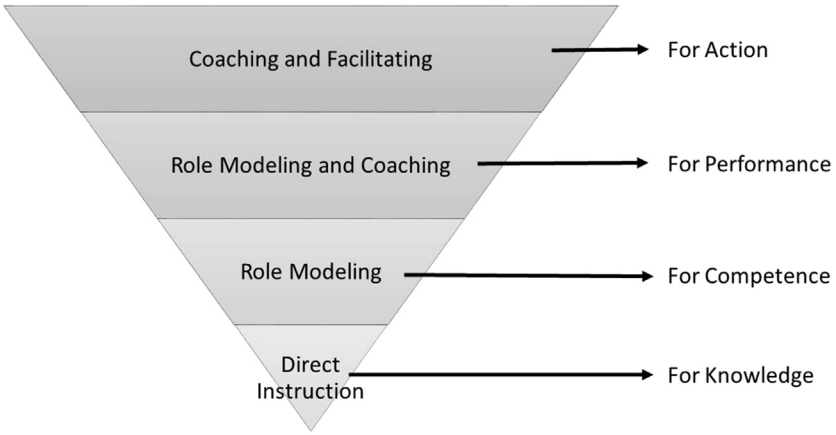


FIGURE 28.2 Roles of a preceptor.[1]

"knows" and "knows how." This knowledge becomes the strong foundation on which experiential learning will build. Correspondingly, preceptors may provide direct instruction only if the foundational knowledge in a specific area is weak.

Further up Miller's pyramid, "shows how" allows the learner to demonstrate what they are able to do. Likewise, preceptors may use modeling to show learners how they want a particular skill carried out. For example, a preceptor may have the student observe a preceptor-to-patient interaction. Once the preceptor feels the student has the knowledge and the ability to perform the skill, the preceptor may utilize a coaching role. The coaching role allows the student to perform the task while a preceptor observes. This allows for confidence building in the student as well as direct and immediate feedback from the preceptor. The topmost point of Miller's pyramid is "does" which corresponds to the preceptor's role of facilitating. When the preceptor serves as a facilitator, the student should be working and learning somewhat independently as the pharmacy laws allow, with the preceptor available as needed.

Often, experiential learning environments have many different levels of learners. For example, there could be an experienced preceptor teaching pharmacy residents as well as students and interns at different levels of their training at the same practice site. This learning model is referred to as layered learning. Layered learning is often used in medical training. The preceptor should clearly delineate the definitions, learning goals, and expectations for each learner in the layered-learning model.

Practice Scenario

You are entering a very specialized field of pharmacy practice for one of your practice-based experiences (e.g., a Pediatric Intensive Care Unit). You have had little to no foundational knowledge in this area prior to your practice-based experience.

- Which role of instruction will your preceptor likely use at the beginning of the experience? *Direct instruction is the likely role the preceptor would use as you may not have a solid foundation in this specialty area of pharmacy practice.*

  Your preceptor asks you to provide education to a patient on the medications while the preceptor observes your performance. Immediately following your patient encounter, the preceptor provides you with direct feedback regarding one strength of your encounter and one area for improvement.
- Which of the four preceptor roles is the preceptor using? *Your preceptor is utilizing the coaching role. The coaching role allows the preceptor to observe and provide immediate instruction and correction as necessary.*

OBJECTIVE 28.2. DESCRIBE THE QUALITIES OF EFFECTIVE PRECEPTORS AND RESPONSIBILITIES OF STUDENTS

Qualities of Effective Preceptors

Professional expertise, empathy, enthusiasm, openness to questions, and giving constructive feedback are just a few of the many attributes that describe effective preceptors. Though awards have been established to recognize preceptors with demonstrated excellence and a long-standing commitment to experiential education, practice, and service,[3] the best judges of qualities of great preceptors, however, are likely students themselves. The Role Model Apperception Tool, designed to assess a clinical trainer as a role model, highlights the importance of both a clinical and caring component of the "3Hs" for positive role modeling: personal (heart), teaching (head), and clinical (hands-on).[4,5] Pharmacy students cite similar role modeling, teaching/coaching, and or facilitating behaviors in award-winning preceptors.[6] A summary of the top qualities cited by learners is found in Table 28.1.

Effective preceptors encourage students to engage fully in learning opportunities to develop their knowledge and skills. Activities should include all those routinely performed by pharmacists, with the level of responsibility commensurate with the student's level of training (beginner, reinforcing, advanced). As identified in Miller's pyramid, the highest level of learning and assessment is to learn by doing. Application of the students' foundational knowledge in real-world patient care settings with other healthcare professional students and practitioners will allow for a smooth transition to practice. Quality preceptors also promote self-directed learning and lifelong learning through the use of strategies such as goal setting, reflection, encouraging creativity in developing new solutions to problems, scholarly inquiry (reading, research, and writing), service-learning, and engagement in activities of leadership and advocacy for the profession.

TABLE 28.1 Selected Qualities of a "good preceptor"

Qualities of Role Model (RoMAT) (Jochemsen) or Modified RoMAT (Bochenek)	Student-Valued Measurable Teaching Behaviors (O'Sullivan)
Clinical ("Hands-On")	**Role Modeling**
Caring Attitude	
• Shows empathy for patients	No equivalent
• Communicates skillfully with patients and their relatives	No equivalent
• Shows enthusiasm for work	• Enthusiasm for patient care
• Interacts in positive way with other healthcare workers	• Works well with other healthcare professionals/colleagues
Effectiveness	
• Possesses excellent clinical reasoning skills	• Very knowledgeable
• Demonstrates professional competence in difficult clinical situations and ability to cope with adversity	No equivalent
Teaching ("Head")	**Teaching/Coaching**
Caring Attitude	
• Shows interest in teaching the rotation[a]	• Teaches students/shares own learning
• Establishes rapport with learners	No equivalent
• Has positive attitude	No equivalent
• Is available to learners	• Goes out of his/her way and always makes time for students
Effectiveness	
• Provides good direction and feedback[a]	• Clear in expectations/provides feedback
• Understands learners' needs and shows commitment to learners' growth	• Interested in student learning and success
• Makes learning stimulating	No equivalent
• Encourages learners to actively participate in discussions and problem-solving exercises[a]	• Answers and asks students questions/has discussions
• Has well-organized and structured rotation[a]	No equivalent
• Is aware of his or her status as role model	No equivalent
Personal ("Heart")	**Role Modeling (Other)**
Caring Attitude	
• Shows patience	• Positive preceptor attitude (e.g., patient, fun, kind)
• Relates to learners as individuals[a]	No equivalent
• Demonstrates honesty and integrity	No equivalent
• Is nice and easy to work with	No equivalent
Effectiveness	
• Is involved with professional pharmacy organizations[a]	• Advocates and has enthusiasm for profession of pharmacy
• Has self-confidence	No equivalent
• Shows leadership qualities	No equivalent
	Facilitating Behaviors
No equivalent	• Flexible, organized, and accommodating to student learning/ scheduling • Establishes great working environment • Solicits student input and suggestions • Willing to provide a variety of experiences

[a]*Not part of original RoMAT instrument. IIIB pharm organization inquiry was not found to be of importance to pharmacy residents.*

Along with recognizing the qualities of an effective preceptor, it is important to recognize the student's responsibilities to ensure that the learning experience is a success. The preceptor will serve as a guide, but the student is ultimately responsible for his or her success. Ongoing communication in the most professional manner possible is key to ensure preceptor expectations are understood. When a student reviews goals with his/her preceptor, learning opportunities can be tailored to meet the needs of the student, preceptor, and practice site to ensure the student can become confident and successful.

Receiving Effective Feedback

One of the greatest responsibilities of a student is to ensure that the preceptor's expectations and feedback are understood throughout the learning experience. One of the biggest mistakes students make is assuming that a lack of negative feedback is a sign that all is going well. Students should ask the preceptor to provide ongoing, routine feedback, as well as a formal midpoint evaluation. The goal is to find out specifically what can be done to improve while there is still time to address concerns. Two types of feedback are highlighted in Table 28.2 and described further below.

Most formal learning experiences include, and most preceptors are quite accustomed to providing feedback that summarizes performance at the end of the experience. This type of feedback is known as summative assessment or summative feedback, which is an assessment *of* learning or measurement of the student's aptitude for accountability purposes. When giving summative feedback, the preceptor takes on the role of an evaluator. This feedback is necessary to determine whether the student is able to progress to the next experience, finish a course of study, or care for patients independently and effectively. The timing of this feedback at the end of an experience, however, does not allow for the student to make modifications to performance that may be below average in real time and learn from their mistakes in a safe environment. A student may be left to discover that they are not performing to a preceptor's satisfaction when it is too late to change the outcome of the experience. This can be very disheartening to the student and quite challenging to the preceptor who does not like to disappoint their trainee.

During the course of the learning experience, there should be many opportunities to learn and to grow in various skill areas. Ideally, the preceptor should proactively offer verbal or written feedback along the way, but if it is not offered, ask for it. This type of feedback, known as formative assessment or formative feedback, is feedback *for* learning, is generally informal, and is intended to serve as a catalyst for further learning. Formative feedback works best when it is (1) embedded in the instructional process and/or workflow, (2) provides specific and actionable feedback, (3) is ongoing, and (4) is timely.[7] Like S.M.A.R.T. (Specific, Measurable, Achievable, Relevant, Timed) goals, a preceptor who functions in the role of a teacher may employ a modified S.M.A.R.T. mnemonic when providing formative feedback to a student: S = Specific, M = Measurable/Meaningful, A = Actionable/Accurate, R = Respectful/Relevant, and T = Timely.[8] By using this approach, discussions involving progress to date, successes, and opportunities for improvement can occur in a timely manner.

Regardless of when feedback is provided, it is important to be open to receiving the feedback; in other words, have a "teachable spirit." As no one is perfect and students are still learning, students should expect to receive some areas of feedback for improvement to allow for growth. When feedback is given that is different from what one might expect, it is important to try to avoid feeling attacked or bad about oneself. Take the opportunity to self-reflect on the performance and be as objective as possible. If questions remain, a discussion to gain more insight with a request for objective examples demonstrating strengths and areas for improvements is recommended. Always be sure to keep the discussion respectful and professional.

Keep in mind the skills learned through the continuing professional development process such as reflection and self-assessment, planning, and evaluation are the foundation for lifelong learning, which is a critical skill for all pharmacists.

TABLE 28.2 Feedback Types During Experiential Rotations

Feedback Type	When?	Goal	Preceptor Role
Formative assessment	Throughout experience	Learning/improvement	Teacher
Summative assessment	End of experience	Measure aptitude/accountability	Evaluator

Practice Scenario

You are scheduled to spend time in a hospital pharmacy for 6 weeks with a preceptor that you do not know much about. You are told to show up each morning on time for rounds and to ask the preceptor if you have any questions. You show up on time for rounds each day, attend required meetings with your preceptor, and turn in your assignments on time. You have been asked questions by physicians and have known the answers about one-half of the time (the other half of the time someone else provided them with an answer). Your preceptor has not provided feedback on your assignments. On the final day of the learning experience, you are told that you received an "average" or "competent" rating on the rotation. You feel confused and upset as you had thought you had done everything you needed to do to get a much better grade on the rotation. When you ask for more specific information, you are told that you needed to come in earlier in the morning to prepare more for rounds and have more information about each patient ready for the team, your knowledge base and responsiveness to questions required more attention, and your writing assignments need improvement.

- What type of feedback did you receive? *In the case above, you received only summative feedback or "assessment of learning." It was provided only at the end of the experience and intended to measure your aptitude in various skill areas.*
- What role did the preceptor play in the above scenario? What is the student's responsibility in acquiring effective feedback? *The preceptor served more as an evaluator throughout this experience than a teacher. Ideally, the preceptor should have provided more clear expectations for you regarding what was expected prior to and during daily patient care rounds. Formative feedback through a variety of mechanisms would be helpful. This can often be accomplished by setting aside even a small amount of time each day for discussion.*

For example, the preceptor might have considered attending rounds with you to provide feedback in real time regarding your skills and contributions to the patient care team. Additionally, having your written work corrected and returned with feedback would have allowed you to view and incorporate suggestions into future assignments.

However, the responsibility does not rest solely with the preceptor to ensure a better outcome for you. You should never assume all is going fine but rather should ask specifically about your performance at minimum once each week. It is best to work with your preceptor to develop a plan for improvement as soon as an issue is identified. In the end, it is important to recognize that until you are able to demonstrate competence, you should not receive a passing grade.

OBJECTIVE 28.3. DETAIL HOW TO DEVELOP A PRODUCTIVE STUDENT–PRECEPTOR RELATIONSHIP

The most productive student–preceptor relationship can be established when both parties actively participate in fostering its development through planning, communication, and the conveyance of respect throughout the spectrum of the experience: prior to, during, at the end of, and following the experience.

Prior to the Experience

Ideally, the preceptor prepares for the student's arrival by completing preceptor training from the pharmacy program, planning a schedule and activities for learning including interactions with as many other healthcare professionals as possible. Preceptors should also acquire space for the student to work and keep belongings, inform other staff of the student's arrival and the role they will play in the student's education, and communicate with the student regarding how to prepare for the experience. At the start of the experience, policies such as arrival time, dress code, and other expectations should be reviewed, leaving no room for ambiguity on the part of the student. Perhaps most importantly, the preceptor should start each experience with renewed passion for teaching and an open mind regarding the student's individual capabilities and learning needs.

The student should also have a solid foundation in understanding the expectations for the learning experience, of the preceptor, and the practice site. As identified by Holt and colleagues, altruism, honesty and integrity, respect for others, professional presence, professional stewardship, and dedication and commitment to excellence are the tenets of professionalism that should be the foundation for every student and pharmacist.[9]

Before each experience, the student should professionally communicate with the preceptor. This communication could include a short introduction and the student's curriculum vitae or resume. Students should reflect on their own strengths and weaknesses while identifying what they hope to learn or gain from the experience. From the very start, the student should have a clear understanding of the schedule and the preceptor's expectations for the experience.

During the Experience

As discussed earlier, the preceptor should strive to provide the student with formative assessment opportunities to offer low-stakes feedback and the opportunity to grow before the final evaluation. When assignments are given, the preceptor should offer explicit instructions to prevent confusion. Additionally, the best preceptor serves as both coach and cheerleader as he/she guides the student through the experience, offering both constructive feedback and praise as needed. Students should accept feedback as a constructive way to improve and not take feedback personally.

The student should be present both physically and mentally each day and should have an organized approach to solving problems. For each problem encountered, students may consider using the SOAP (Subjective, Objective, Assessment, Plan) approach to thinking through the problem. Subjective information includes the observations or data the patient shares with the pharmacist. For example, it may include symptoms or feelings expressed by the patient or the person asking the question. Objective information is data that can be measured or quantified. This may include laboratory values or numbers that have been quantified related to the problem. Assessment is a way to state the problem that was identified, in other words, the diagnosis of what is wrong. Finally, the plan is the student's recommendation to solve the problem. Using the SOAP approach is a quick way to communicate an issue with a preceptor whether it is written or verbal communication. A newer, but similar method to SOAP is pharmacists' patient care process (PPCP).

The preceptor is present to assist with learning and reasoning through problems appropriately. It is also of the utmost importance to receive feedback during the experience to make performance improvements and ultimately learn. If a preceptor does not provide feedback, ask for it in a way that is positive. Such as, "I would really appreciate your advice on how I can improve in this activity." In addition, every patient deserves the best performance in every encounter, regardless of the situation. As a student and learner, it is ok to say, "I don't know, let me look it up and get back to you." It is never ok to guess when it comes to caring for a patient.

During the practice experience, students may be introduced to multiple members of an interdisciplinary or interprofessional healthcare team. Interprofessional education occurs when two or more professionals learn with, from, and about each other to allow for collaboration and improved health outcomes for patients.[10] Interprofessional education and practice involves students and preceptors in exhibiting competency in having mutual respect and shared values of other professions, knowledge of roles of each health professional, communication with patients, families, communities, and other professionals (both health and other fields), and team dynamics and relationship-building skills to care for patients.[11] Students should clearly understand the role of each individual of a healthcare team and take time to develop communication styles and team building skills during each practice-based experience to embody the competencies of interprofessional education and practice.

At the End of the Experience/Following the Experience

At the conclusion of the experience, the preceptor will have feedback for the student regarding their overall performance. As part of this evaluation, students may be asked to complete a self-assessment as a way to create a dialog back and forth between preceptor and student, which serves as an opportunity to strengthen the quality of the relationship. If both the preceptor and the student have been proactive throughout the rotation in fostering the relationship, nothing said during the final evaluation will be a surprise. If a surprise does reveal itself, let it serve as a valuable lesson regarding the importance of soliciting feedback from the preceptor throughout the experience.

Lastly, it is important to recognize what to do after the practice-based learning has occurred through gratitude and reflection. Gratitude practices allow the student to show appreciation to his/her teacher. One of the recommended ways this can be done is through a simple handwritten note of thanks. Reflecting and documenting achievements during the experience or journaling is a way to reflect on what has been learned and to track meaningful encounters with patients, preceptors, and other healthcare providers. Kolb describes a method involving reflection in the experiential learning cycles. It can begin with describing what happened in an experience. The next step is an analysis or reflective observation of what happened, including the meaning this experience had on the students' learning. Making generalizations about the experience is the next step. Finally, the student identifies future action through planning and the experiential learning cycle continues.[12]

Ready for Practice

The critical outcome for all experiential learning is creating students who are ready to enter professional practice as a pharmacist (i.e., practice-ready), prepared to work on a team (i.e., team-ready), and ready and willing to teach the next

generation of pharmacists (i.e., preceptor-ready). Practice-ready and team-ready graduating students are competent to contribute to patient care by working in collaboration with other healthcare providers. Maintaining a forward-thinking, visionary approach to learning and development will ensure that you are ready to practice the profession of pharmacy and become a future preceptor on graduation.

Practice Scenario

You are about to have your first practice-based experience in a hospital setting. Before the experience, you researched information about the hospital and communicated with the preceptor regarding where to report on the first day.

- What steps did this student miss in preparing for a practice-based experience? *One critical step missed was to self-assess and identify your strengths and weaknesses. A second step in self-assessment is to plan and identify what you would like to learn throughout the experience as it relates to this specific practice area.*

CONCLUSION

What to expect from a preceptor is different from what is expected from faculty teaching in a classroom. How to adapt to experiential education sites, preceptors, and the healthcare team is to be planned well in advance. Preceptor roles vary depending on where the student stands on performance level for a particular task. Understanding the roles and responsibilities of the preceptors and interns is pivotal in establishing a productive relationship.